

Data Quality Report — Q2 (Real Data)

One page. Written form of the Q2 answer on the **real** 10,547-review corpus. Full JSON: data/processed/defect_detection_report_real.json. Run standalone: python3 src/real/detect_defects_real.py.

Product-inclusion quality (a real-data problem a synthetic fixture cannot have)

Detection. "Appliances" metadata yielded only 32 candidates, dominated by replacement-filter accessories misclassified as purifiers. Switched to "Home_and_Kitchen"; a keyword classifier there produced real false positives — vacuum cleaners, wearable necklace ionizers, home-decor products whose *description* mentioned "air purifier" in cross-sell copy.

Remedy. Tightened to title-only matching with regex exclusions, then a minimum 5-real-rating threshold. Candidate count moved 32→106→227→**237 final**, re-inspected by hand at each step.

Effect of not catching it. The corpus would have included hundreds of replacement-filter and vacuum-cleaner reviews under an "air purifier" label, corrupting every downstream prevalence and CSAT figure.

Defect: rating/text sentiment contradictions

Detection. Negation-aware polarity scoring (rating==5 with strongly negative text, or rating==1 with strongly positive text).

Detected: 239 / 10,547 (2.27%).

Remedy. Quarantined (rating_trusted=false), not deleted — the text is real experience, only the star label is untrusted.

Effect of not catching it, quantified via a real detector bug. The first version had no negation handling and misread "I cannot recommend this product" as *positive*. On a partial run this produced 248 inflated flags; adding an 18-character negation window dropped it to 186, and a wider, real-vocabulary term list landed at 239 on the full corpus. **A downstream metric on the un-fixed version would have systematically overstated how often 5-star reviews hide complaints.**

Defect: empty/trivial review text

Detection. len(review_text.strip()) < 3 characters.

Detected: 18 / 10,547 (0.17%).

Remedy. Removed — no analyzable signal.

Non-defects (real, honest findings)

0 exact-duplicate-text bursts (≥3× on the same product) and ****0 per-product daily volume anomalies**** (robust z-score ≥5). A genuine result, not a detector failure — absence of a defect is itself a finding worth stating plainly.

Summary

Defect	Count / Corpus	Remedy
Product misclassification (pre-corpus)	227 dropped from 227 v1 candidates during 2 validation passes	classifier tightened, re-inspected by hand
Sentiment/rating conflicts	239 / 10,547 (2.27%)	quarantined (rating_trusted=false)
Empty/trivial text	18 / 10,547 (0.17%)	removed
Duplicate-text bursts	0 (real finding)	none needed
Volume anomalies	0 (real finding)	none needed